# Design of Scheduling Algorithms for Vehicular Cloud Computing

*Seminar / Project report submitted to the Biju Patnaik University of Technology*

*for the partial fulfillment for the award of the degree of*

**Bachelor of Technology**

*in*

**Computer Science and Engineering**

*by*

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25/06/2021

# CERTIFICATE

This is to certify that the work contained in the thesis/project report entitled **"Design of Scheduling Algorithms for Vehicular Cloud Computing"**, submitted by **Student name (Regd. No.: XXXXXXXXX)** for the award of the degree of **Bachelor of Technology** to the **Biju Patnaik University of Technology, Rourkela, Odisha** is a record of bonafide research works carried out by him/her under my direct supervision and guidance.

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| Dept. of CSE, SIT | | |

# Declaration

I certify that

(a) The work contained in the thesis is original and has been done by myself under the supervision of my supervisors.

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# Abstract

In the last decade, vehicular cloud computing has received huge attention in business and scientific communities. The domain of vehicular cloud computing integrates two emerging fields, namely cloud computing and vehicular ad hoc networks.

***Keywords:*** Cloud Computing, Vehicular Ad Hoc Network,

# Acronyms

| | |
|---|---|
| CaaS | Computing as a Service |
| COaaS | COoperation as a Service |

# Notations

| | |
|---|---|
| $A$ | A large urban area |
| $ACOST[j]$ | Promised cost to execute UR $U_j$ |

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# Chapter 1

# Introduction

## 1.1 Motivation

Rapid evolution of cloud computing paradigm provides on-demand service provisioning of resources, such as computing, storage, network and applications on a pay-as-you-use basis [1]. The architecture of cloud computing is shown in Figure 1.1


Figure 1.1: Architecture of cloud computing

The data are shown in Table 1.1

Table 1.1: Data for Cloud

| One | Two | three |
|---|---|---|
| 1 | 2 | 3 |
| One | Thrre | Four |

# Chapter 2

# Literature Review

## 2.1 Introduction

The pioneering advancement in vehicular technology introduces VCs to the business and scientific communities. VCs open up a large number of opportunities and challenges to the researchers

# Chapter 3

# Dynamic Service Migration Algorithm

## 3.1 Introduction

The mapping and migration problems of the URs to the VMs are discussed in this chapter. Here, we explore the mapping of $l$ URs to the VMs hosted by $m$ vehicles in a large urban area with g grids and present an algorithm called dynamic service migration (DSM) for vehicular clouds. The proposed algorithm's objectives are: minimize the number of migrated URs and dropped URs and maximize the number of completed URs.

# Chapter 4

# Smart Cloud Service Management Algorithm

## 4.1 Introduction

In this chapter, we consider a large area with $g$ grids and two sets containing $l$ number of URs and $m$ number of vehicles/VMs, respectively. Here, we formulate three sub-problems, namely mapping vehicles to grids, URs to grids and URs to vehicles, and present an algorithm called SCSM for VCs to address various challenges, discussed in Chapter 1. The objectives are to minimize makespan ($M$), number of migrations ($NoM$), average turnaround time ($ATAT$), average waiting time ($AWT$), average response time ($ART$) and average idle time ($AIT$), and maximize average resource utilization ($ARU$).

# Chapter 5

# Resource Management Algorithm for VM Migration

## 5.1 Introduction

In the present chapter, we come up with a novel resource management algorithm, called resource utilization-aware VM migration (RU-VMM), to distribute the loads among the overloaded and underloaded vehicles. Here, we consider a set having $m$ number of vehicles along with a matrix $VMU$. The goal is to reduce the number of source vehicles for the VM migration process, minimize the percentage of dropped migration, and maximize the percentage of successful VM migration. Here, the $VMU$ matrix shows the utilization of the VMs hosted on different vehicles.

# Chapter 6

# Energy-Aware and Revenue-Based Service Management Algorithms

## 6.1 Introduction

An energy-aware service management (EASM) algorithm is introduced in this chapter that considers properties of GVs and CVs along with the properties of URs to provide services. EASM assigns a set containing $l$ number of URs to a set containing $m$ number of vehicles. The key focus of EASM is to complete more number of high priority URs with minimum cost. The proposed algorithm also maximizes the profit of the CSP.

# Chapter 7

# Conclusions and Future Scope

## 7.1 Conclusions

Vehicular cloud computing has earned remarkable recognition in the recent era. In our work, we have inspected various aspects of VCC along with its emergence of popularity, its applications across different dimensions of this tech-driven world, and several research challenges posed by it. We have proposed various UR based scheduling algorithms by considering numerous distinct factors like completion time, resource utilization, makespan, energy consumption, renewable energy utilization, revenue, price and many more.

# References

[1] Rajkumar Buyya, Chee Shin Yeo, Srikumar Venugopal, James Broberg, and Ivona Brandic. Cloud computing and emerging it platforms: Vision, hype, and reality for delivering computing as the 5th utility. *Future Generation computer systems*, 25(6):599–616, 2009.

# Dissemination

## List of International Journals

1. **Sohan Kumar Pande** et al., “Dynamic service migration and resource management for vehicular clouds”, **Journal of Ambient Intelligence and Humanized Computing**, Springer, pp. 1-21, 2020 (**Impact factor: 4.594, SCI & SCOPUS Indexed**).

## List of International Conferences

1. Sohan Kumar Pande et al., “A Revenue-Based Service Management Algorithm for Vehicular Cloud Computing”, $17^{th}$ **International Conference on Distributed Computing and Internet Technology**, Springer, Lecture Notes in Computer Science (LNCS) Book Series, Vol. 12582, pp. 98-113, 2020.

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ORIGINAL RESEARCH

# Dynamic service migration and resource management for vehicular clouds

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**Abstract**

Vehicular cloud computing has received huge attention in business and scientific communities, which integrates two emerging fields, namely cloud computing and vehicular ad hoc networks. It acts as a data center by using the underutilized resources of the networked vehicles. Moreover, many studies suggest these vehicles as potential candidates for hosting virtual machines (VMs). As a result, a VM can be set up using the vehicular resources, and it can be transferred from one vehicle to another vehicle to continue its execution, under some circumstances. It enables the vehicular environment to provide services to the user requests, that are submitted to the cloud. However, the mapping of such requests to the VMs (or hosted vehicles) and service migration is very much challenging, and not well-studied in the literature. In this paper, we propose a dynamic service migration algorithm for vehicular clouds. The algorithm consists of three phases, estimation, assignment and migration. The performance is carried out through simulation runs using two scenarios of six datasets, and compared with three well-known algorithms, namely vehicular VM migration-uniform, round robin and mobility and destination workload aware migration using four performance measures. The comparison results followed by statistical validation using *T* test show the superiority of the proposed algorithm over the existing algorithms.

**Keywords** Cloud computing · Vehicular ad hoc network · Vehicular cloud · Migration · Resource management · Roadside unit

## 1 Introduction

The emergence of cloud computing led to the provisioning of the computing facilities and resources as services to the users in an on-demand and pay-per-use basis (Buyya et al. 2009; Panda and Jana 2015; Pande et al. 2016; Wei et al. 2010a, 2014, 2010b). These services are rendered in the form of virtual machines (VMs) that become a part of physical resources (or data centers), which increases the scalability. Alternatively, a user can set up the VM environment, without concerning the underlying cloud infrastructure (Refaat et al. 2016; Panda and Jana 2018; Panda et al. 2018; Rahimi et al. 2014; Mastelic et al. 2015). The cloud computing services have drastically increased and covered a wide range of possibilities with the integration of devices via the Internet. Here, the devices are not limited to tablets, personal computers, laptops and personal digital assistants, they also include smart vehicles, smart gadgets and many more. As a result, cloud computing is more appealing in business and scientific communities (Panda et al. 2019; Panda and Jana 2019a, b).

In recent years, vehicular networks have gained immense popularity based on the increasing involvement of smart vehicles. A smart vehicle is equipped with an on-board unit, which consists of processing unit, storage unit, global positioning system, radar device, sensors, communication devices, digital map, cameras and many more (Kim et al.

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